r/NoStupidQuestions

How bad is AI really for the environment?

Posted by u/wholenewguy • • 0 points • 9 comments

I've been hearing more and more about this, and it does make sense considering the power and water to drive this much computing must come from somewhere. But I just don't know

Comments

u/Bob_NotMyRealName • 1 points • 2024-10-08 16:42 UTC

Creating 1,000 images with a model like Stable Diffusion XL produces as much carbon dioxide as driving just over four miles in a gas-powered car. Now imagine a million people creating images, and that's only a very tiny sample of what AI can/is doing.

So yeah, it's significant.

u/AI_optimist • 1 points • 2024-10-08 16:41 UTC

Training AI takes a lot of electricity to power the datacenters, and the cooling systems for those centers. If the area those data centers are located in rely on fossil fuels, then it can be bad for the environment. It's worth noting that the methods to train AI were discovered at a point when nobody was making specialized energy efficient AI compute hardware. Now people know that we do need energy efficiency and there are tons of companies working on it.

So the energy issue for training is temporary...

When AI is generating something it uses very little energy. A text output uses about the energy as a google search.

An image generation uses about as much power as playing 1 second of a modern videogame.

> **u/Delehal** • 1 points • 2024-10-08 16:45 UTC
>
> >the energy issue for training is temporary
>
> I'm not so sure about that. New generative AI models are using more power than the older ones. The power requirements seem to be increasing exponentially over time. If we make the chips more efficient, companies will make and use more chips. Executives at AI companies have been pretty clear that future AI products will use even more power than current ones.
>
> > **u/AI_optimist** • 1 points • 2024-10-08 17:01 UTC
> >
> > >I'm not so sure about that

I am 110% sure about it.

>New generative AI models are using more power than the older ones.

Yes, that is why I compared the power output to modern videogames instead of videogames that are older and use less energy to process.

>The power requirements seem to be increasing exponentially over time.

The exponential power increases is for training AI. Once it is trained, there are not noticeable differences in energy usage.

The trained model doesn't get more intensive to run. The Image gen I would run on my computer 2 years ago uses the same resources as the most recent ones like Flux. What changed is the quality of the AI model, not the amount of math the computer needs to do to computer the image.

I know conventional logic might seem like "more details means more energy per generation" and that is not the case.

u/Delehal • 0 points • 2024-10-08 17:20 UTC

>I am 110% sure about it.

I was trying to be polite. You are flat out wrong I'm afraid.

>The exponential power increases is for training AI. Once it is trained, there are not noticeable differences in energy usage.

This is wrong in two important ways. First, companies don't just train their models once and then stop. Second, AI power usage has continued to climb over time. The data just is not on your side here.

Is there any credible expert in the field that thinks AI power usage is going to trend downward over time? I just have no idea where you're getting this from.

u/AI_optimist • 1 points • 2024-10-08 17:33 UTC

I could share things that are credible, but you've already demonstrated that you're willing to bend the discussion towards your opinion.

> you: "If we make the chips more efficient, companies will make and use more chips."

You almost admit that chips will get more efficient, but then you make up artificial demand for them that would for some reason negate the benefits..?

I'll be talking about energy efficiencies getting better (not to mention renewable energy) and how that helps the "energy issue" and you choose to interpret that as

"**AI power usage is going to trend downward over time**". Not only did I never say that, it makes discussion impossible if you're going to invent inferences.

I have faith we're both confident in our assessments of AI and it's power consumption. Mine is from first hand use and yours is from wherever it's from. How about we meet up back here in 4 years and see which way it ended up trending towards :)

u/XylitolCarpet • 1 points • 2024-10-08 16:39 UTC

Not as bad as show manufacturing

u/ImpedeNot • 1 points • 2024-10-08 16:35 UTC

AI datacenters are projected to make up 6% of the total electrical consumption in the US by 2026, so yeah, it's significant.

Clean power would reduce that significantly.

u/BobIsMyCableGuy • 0 points • 2024-10-08 16:25 UTC

It's about as bad as a car's emissions. Maybe slightly higher, but not by much.

In all honesty, cars are worse for the environment (whether they be electric, or gas powered.) So are Factories. Plenty of computers/servers have become naturally better at being energy efficient as they have been developed so in general, your computer will still use less power than your Refrigerator. The problem isn't really the AI itself, but the appliances that need to keep the servers/computers running the AI cool.